

Davis Rattनाविजै

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Portfolio: <https://tinyurl.com/davisrattनाविजै-portfolio>

EDUCATION

Cornell University, College of Engineering, Ithaca, New York Expected May 2028
Bachelor of Science in Mechanical Engineering (Intended), Minor in Electrical and Computer Engineering (Intended)
GPA: 3.85 | Major GPA: 3.91
Coursework: Calculus III, Linear Algebra, Differential Equations, Physics Mechanics and E&M, Statics and Mechanics of Solids

AWARDS

- Modeling the Future Challenge (MTFC) 3rd Place, 2024: Won \$10,000 and ranked 3/268 in actuarial science competition
- International Mathematical Modeling Contest (IMMC) USA Regional Finalist, 2024: Ranked in the top 5 paper submissions
- High School Mathematical Contest in Modeling (HiMCM) Finalist, 2023: Ranked in the top 6% of 967 paper submissions

EXPERIENCE

Suspension Team Member and Purchasing Coordinator Feb 2025 – Present
Cornell Formula SAE Racing, Cornell University, Ithaca, NY

- Collaborate with 60+ engineers to design and build a formula-style electric race car for the annual Formula SAE competition
- Design and optimize a custom welding jig for suspension links reducing jig components by 28%, size by 26%, and build time
- Work closely with suspension link designer several times a week, reviewing design choices, load calculations, and CAD
- Present and defend technical decisions in multi-stage design reviews through mock-ups, slide decks, and written reports
- Read literature and work with senior team members to gain system-level understanding in Vehicle Dynamics
- Manage the purchasing and receiving team, handling approximately 1,200 orders annually across all engineering sub-teams
- Develop new features for the team's internal purchasing workflow using Google Apps Script for automation

Seasonal Teller May 2025 – Aug 2025
Middlesex Savings Bank, Wayland, MA

- Worked closely with managers, bankers, and tellers during fast-paced 44-hour weeks to run the bank's busiest branch
- Delivered customer service in-branch and over the phone to hundreds of clients (e.g. new customers, veterans, and businesses)
- Processed thousands of customer transactions, including deposits, withdrawals, loan payments, check cashing, and CDs
- Ensured day-to-day compliance with identity verification, dual control, audits, fraud detection, and documentation procedures
- Certified in financial regulations (e.g. KYC, BSA, AML); maintained strong knowledge of bank policies, products, fees

Project Lead Sep 2024 – Jan 2025
Cornell Data Journalism, Cornell University, Ithaca, NY

- Led a team of 5 analysts to research, analyze data, and publish 3,000-word article on Social Security Trust Fund Insolvency
- Analyzed socioeconomic trends, Social Security data, and policy proposals with Python to evaluate future outlook
- Presented technical updates to 50+ analysts; collaborated with the executive board and editorial team on publication

Founder and Competitor Nov 2021 – May 2024
Math Modeling Team, Newton North High School, Newton, MA

- Founded and led a 4-person team in 7 national/international math modeling competitions (HiMCM $\times 3$, MTFC $\times 2$, IMMC $\times 2$)
- Co-authored seven 25-page math modeling papers; presented one at the 2023 Joint Mathematics Meeting in Boston
- Built models with regressions (linear, non-linear, symbolic), time-series forecasting, cosine similarity, and Lagrange multiplier
- Applied Python, R, MATLAB, and Excel for model validation, statistical analysis, and data visualization for all math models

SKILLS

General: Teamwork, Problem Solving, Communication, Customer Service, Data Analysis, Mathematical Modeling

Software: Python, MATLAB, Autodesk Inventor, Microsoft Word, Microsoft Excel

Languages: English, Mandarin (Conversational), Japanese (Limited Proficiency)

ACTIVITIES & INTERESTS

Intramural Recreational Soccer, Skiing, Biking, Personal Investing, Mahjong, Dragon Boat Steering, Cardboard Model Building